26 August 2026

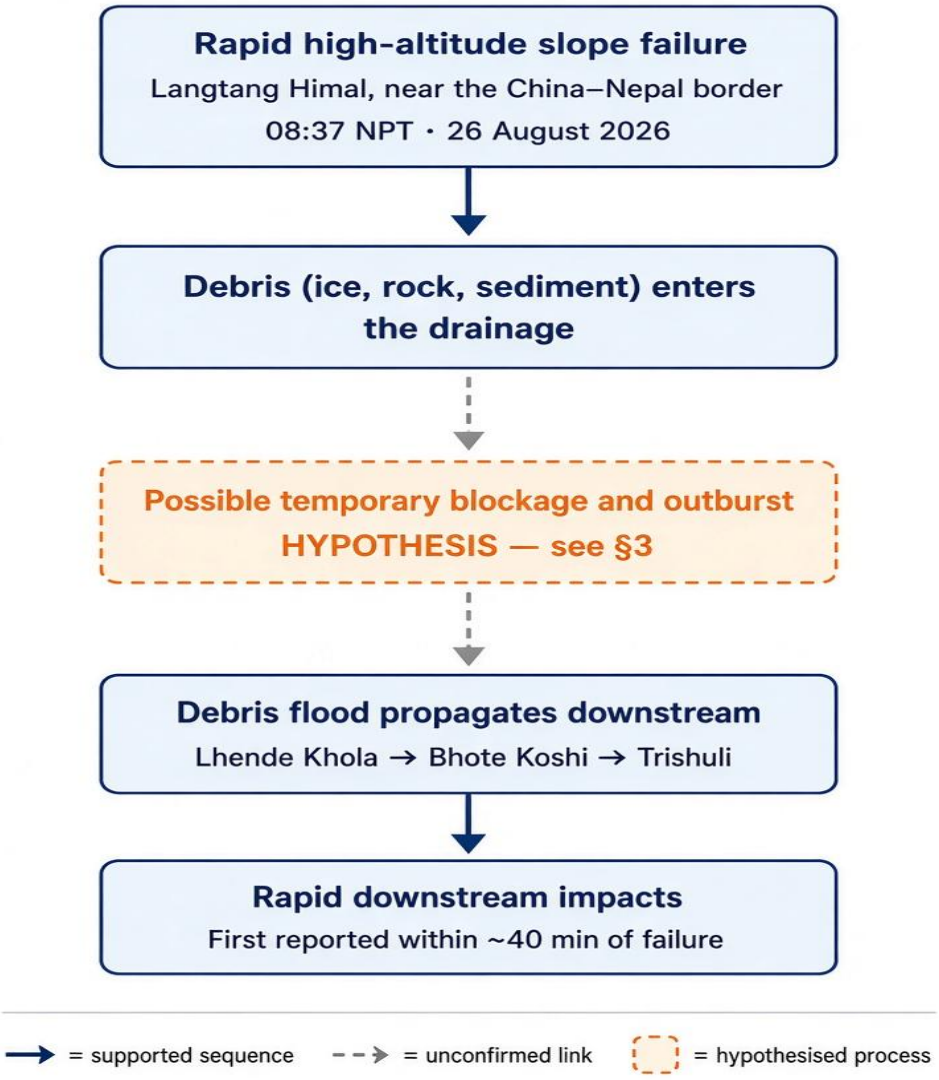
High-altitude slope failure and debris flood

Bhote Koshi–Trishuli corridor

Event reconstruction and illustrative loss workflow on public data

Hazard chain: supported sequence vs hypothesised blockage

- **Supported sequence:** slope failure → debris enters drainage → debris flood downstream → rapid impacts
- Temporary blockage and outburst remains **HYPOTHESIS**
- Source peak (Langtang Lirung vs Tsangbu Ri) and volume estimates remain unresolved
- Diagram distinguishes evidence status from the physical sequence



Why the corridor is high-risk: concentrated infrastructure and interconnected losses

- **Narrow valley corridor:** border crossing, roads, bridges, settlements, and hydropower projects share the same path.
- **Official figures (Rasuwa):** 41 bridges swept away + 4 damaged; ~41 km of road damaged/destroyed; 14 hydropower projects (~748 MW combined).
- **Syabrubesi:** >240 buildings destroyed, 32 damaged (Copernicus EMSR927 grading, 27 Aug 2026).
- **Full asset-level intersection** with flood depth was not modelled (no inventory + no event-specific depth layer).



Pathway and selected hydropower sites; full road/bridge/settlement layers not shown (no complete inventory in this case study).

Insurance market context: NIA claims snapshots and reporting dates

- **Two dated NIA snapshots** (do not blend): distinct reporting windows and criteria.
- Figures are **filed claims**, not paid losses.
- Claims are heavily concentrated in **commercial and hydropower assets**.

Snapshot Date	31 Aug 2026	29 Sep 2026
Claims Filed	**NPR 25.87 bn**	**NPR 20.98 bn**
Policies	583 policies	2,180 policies
Terms & Status	Initial snapshot	Claim deadlines waived; advances up to half of prelim survey

Source: Nepal Insurance Authority snapshots compiled in the project's insurance layer.

Insurance claims vs reconstruction need: the protection-gap question

- **31 Aug snapshot:** NPR 25.87 billion filed (583 policies)
- **Reconstruction need** is substantially larger
- **Filed ≠ paid:** reconstruction need is broader than insured loss
- **Difference** indicates a gap, not a precise uninsured-loss figure

31 Aug filed claims	**NPR 25.87 bn** (583 policies)
Reconstruction need range	substantially larger
Status	filed ≠ paid

** 29 Sep snapshot recorded separately; not blended.*

Illustrative model output vs reported national loss: US$33M ≠ US$2.6B

- **Model output on synthetic assets:** ~US$33.3 million ground-up loss
- **National-loss estimate:** ~US$2.6 billion
- **Different scope:** toy inventory vs national aggregate
- **Vulnerability functions** are JRC-informed placeholders, not Nepal-calibrated
- **Do not retune curves** to force agreement

Illustrative model	**~US$33.3 million**
National estimate	**~US$2.6 billion**
Difference	Scope + data limits

Risk-financing layers: who bears losses that insurance does not cover?

- **Insurance covers only part** of commercial and hydropower losses (filed claims are not paid losses)
- **Remaining losses** fall on asset owners, lenders, and government budgets
- **Reconstruction need** is broader than insured loss
- **No company-level payout data** or donor ledger in this case study

Insurance Layer	Filed claims (partial, concentrated)
Private Layer	Asset owners and lenders
Public Layer	Government and reconstruction finance

Two major events in 14 months: what does this mean for return-period assumptions?

- **July 2025:** Rasuwagadhi Glacial Lake Outburst Flood (GLOF) (Tibet-side source)
- **August 2026:** high-altitude slope failure (Nepal-side)
- **Same general corridor,** ~14 months apart
- **Two events** do not establish a 14-month return period
- **Raises a question** about clustered events and conventional assumptions

Prior Event	**July 2025 Rasuwagadhi GLOF**
This Event	**August 2026 slope failure**
Implication	Questions for return-period assumptions (not a new frequency claim)